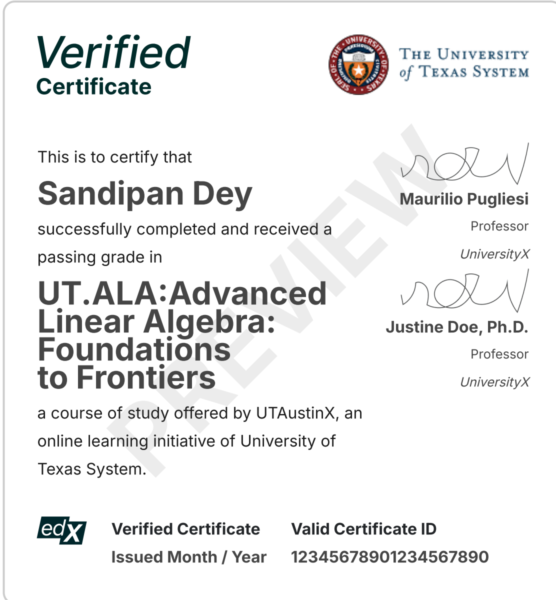
1.3.6 Computing the matrix 1-norm and inf-norm

Bookmark this page

Pursue a verified certificate



- Unlimited access to all course materials
- Shareable verified certificate
- AI learning assistant and translations

Upgrade for $99

Week 01 due Jan 9, 2025 09:15 IST

Video

measures

how much the linear transformation stretches using the 1-norm for the input

and for the output. Now let's look at that matrix infinity-norm.

Defining it is easy., Just replace everything with the Infinity norm. And now

let's move over here and ask the question, "What is the infinity norm of

this matrix?" Well, turns out again there is a simple formula for this. You look at

the vector 1-norm of the row. So you think this as a column vector. You take

its 1-norm. The 1-norm

3:40 / 3:40 2.0x

Video

Download video file

Transcripts

Download SubRip (.srt) file

Download Text (.txt) file

Reading assignment

1/1 point (graded)

Read Unit 1.3.6 of the notes. [LINK]

☑ done

Submit

Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 01 / 01.3.6

Add a Post

Show all posts by recent activity

? Infinity Norm 7

In the lecture, when we are considering Infinity norm, as we suppose to consider column vec...

Homework 1.3.6.1

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$ and partition $A = \left(\begin{array}{c|c|c|c} a_0 & a_1 & \cdots & a_{n-1} \end{array} \right)$.

$\|A\|_1 = \max_{0 \leq j < n} \|a_j\|_1$.

ALWAYS ✔ Answer: ALWAYS

For full solution see notes: [LINK]

Submit

Answers are displayed within the problem

Homework 1.3.6.2

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$ and partition $A = \left(\begin{array}{c} \tilde{a}_0^T \\ \hline \tilde{a}_1^T \\ \hline \vdots \\ \hline \tilde{a}_{m-1}^T \end{array} \right)$.

$$\|A\|_\infty = \max_{0 \leq i < m} \|\tilde{a}_i\|_1 \; (= \max_{0 \leq i < m} (|\alpha_{i,0}| + |\alpha_{i,1}| + \cdots + |\alpha_{i,n-1}|))$$

Notice that in this exercise $\tilde{a}_i$ is really $(\tilde{a}_i^T)^T$ since $\tilde{a}_i^T$ is the label for the ith row of matrix A.

ALWAYS ✔

Submit

✔ Correct (1/1 point)

‹ Previous Next ›

edX

About
Affiliates
edX for Business
Open edX
Careers
News

Legal

Terms of Service & Honor Code

Privacy Policy
Accessibility Policy
Trademark Policy
Sitemap
Cookie Policy
Your Privacy Choices

Connect

Idea Hub
Contact Us
Help Center
Security
Media Kit

GET IT ON Google play
Download on the App Store